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A Theory of Auditor Resignation

KEITH BOCKUS AND FRANK GIGLER*

1. *Introduction*

In this paper we propose an economic rationale for auditors resigning from the engagements of "risky" clients and use this rationale to examine claims that increases in expected auditor liability are giving rise to an increase in auditor resignations.[1] Given the relatively inelastic demand by publicly traded firms for audits, it seems puzzling that auditors would resign rather than simply risk-adjust their bids for the increase in expected liability. In addition, it is not clear why a *successor* auditor, who faces not only the threat of legal liability but also the information disadvantage accompanying a first-year audit, would accept the rejected client. Our theory explains how an incumbent auditor rationally resigns an engagement because any attempt at risk-adjusted pricing leaves him with only unprofitable clients, and how successor auditors who know less about the client than the incumbent auditor can profitably accept the engagement of a client whose auditor has rationally resigned.

Several auditing models have specifically addressed why client firms switch auditors (Magee and Tseng [1990], Dye [1991], Teoh [1992], Kanodia and Mukherji [1994], and Gigler and Penno [1995]). The interest in auditor switches arises partly from regulatory concerns that

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[1] "More Accounting Firms Are Dumping Risky Clients," *Wall Street Journal* (April 25, 1997), and Holland, Galen, and Light [1993].

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auditor switching might compromise financial reporting (opinion shopping) and partly from interest in how capital markets interpret auditor switches. Originally, audit pricing models (such as DeAngelo [1981; 1982]) did not address switching explicitly because their assumption of identical auditors ruled out any incentive for auditor switches. More recent models all introduce some sort of difference across auditors to create a demand for auditor switches.

We use both an information difference and an exogenous difference across incumbent and successor auditors to generate auditor switches. Specifically, we assume the auditor is unsure about whether his client has a hidden litigation risk. Incumbent auditors have better information about this partial risk than do possible successor auditors. Even so, incumbents may be at a cost disadvantage, relative to a potential successor auditor, simply because they face higher liability than the potential successors. For the most part, our analysis assumes that differences in auditor wealth imply differences in liability. If some auditors face less liability than the incumbent because they are more wealth constrained, and if the expected liability to the incumbent auditor is sufficiently high, any attempt by the incumbent to cover his expected liability with the audit fee will drive off a good client to a successor auditor, while only a bad client has an incentive to retain the cost-disadvantaged incumbent auditor. This adverse selection problem faced by the incumbent auditor may make it impossible for him to post a viable risk-adjusted bid, leaving him with no choice but to resign. The successor, by contrast, engages the *pool* of clients following a resignation, thereby making it profitable for him to accept the engagement of a client whose auditor has resigned.

The insight that price may adversely sort potential customers is illustrated in a credit market by Stiglitz and Weiss [1981]. In their model, banks set interest rates too low to equate the supply and demand for funds because raising the rate merely encourages the less risky borrowers to self-finance. Our setting differs from Stiglitz and Weiss [1981] because we explicitly model competition among differentially informed heterogeneous producers. Consequently, our results differ qualitatively in that our incumbent auditor cannot protect himself from the adverse selection by lowering his price, giving him no alternative to resignation.

Auditor resignations are observed in our model only if the incumbent auditor assesses a sufficiently high probability that his client presents a significant litigation risk. This may be a rare condition. Defond, Ettredge, and Smith [1992] show that approximately 6% of auditor switches are reported as either auditor resignations or declinations to stand for reelection and that about half these instances are either not explained or are explained as resignations for cause (that is, a resignation for reasons specific to the client, the others being some auditor-related event which makes it impossible for the auditor to continue to serve the client). Even though resignations for cause are rare, they are economically significant; Defond, Ettredge, and Smith find that firms whose auditors resign for cause suffer significant abnormal returns of −11.8%.

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Consistent with practitioner claims, the comparative statics of our model reveal that increasing auditor liability increases the frequency of auditor resignations and that resigned clients seek audits from smaller (i.e., wealth-constrained) auditors. Furthermore, these clients receive less demanding audits from their successor auditors.

Section 2 presents the model. Section 3 characterizes the equilibrium. Section 4 gives the comparative statics and section 5 concludes the paper with a discussion of its potential limitations and empirical implications. All proofs are contained in Appendix A.

2. *The Model*

2.1 SETTING

We consider a two-period model with multiple auditors competing for the audit of a single representative client. The client privately knows whether he presents a significant risk to the auditor; it is the auditor's job to detect any such risk. If the auditor detects the risk, the client will incur an incremental expected cost (or penalty in the case of a fraud) of P. An auditor who fails to detect the risk will incur an incremental expected liability of L. The liability and the penalty establish natural incentives for auditors to detect any hidden risk and for clients with hidden risks to try to escape detection.[2] An extreme example of such a risk would be management fraud. But, all we require is that the client privately knows of the risk (and does not want it uncovered) and the auditor wants to uncover the risk.

Auditors differ in their abilities to absorb liability. That is, auditors with wealth $W \leq L$ face an incremental expected loss of W for failing to detect a hidden risk. These wealth-constrained auditors are viewed as lower-quality auditors.[3] The client receives a benefit, B, from hiring an auditor who is not wealth constrained (hereafter a "solvent" auditor). Let the set of all auditors be denoted $\mathbf{A}$, the subset of wealth constrained auditors be denoted $\mathbf{\Omega}$, and the subset of solvent auditors be $\mathbf{\Sigma}$. To avoid settings where a client and an auditor can benefit by contracting to have the auditor ignore a risk, we assume $L \geq W \geq P$.

Let Δ_i be the probability of auditor i detecting a risk given that the client has a hidden risk, and let the probability of wrongly detecting a nonexistent risk be zero for all auditors. Δ_i is chosen by auditor i after he engages the client and learns whether the client's previous auditor has resigned. The cost of choosing a probability of detection is represented

[2] The adversarial nature of the audit relationship distinguishes it from other professional/client relationships.

[3] This will be borne out in equilibrium, as we shall see that wealth-constrained auditors, having less to lose, will be less likely to detect a risk. Conversely, one can think of the auditor's quality as exogenous and the liability endogenous. High-quality auditors have more to lose from litigation because of the rents generated by B, their reputation capital. Hence, they will work harder than a low-quality auditor to protect those rents.

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by the cost function $C(\Delta)$, with the following properties: $C'(\Delta) > 0$ for all $\Delta > 0$, $C'(0) = 0$, $C'(\Delta) \to \infty$ as $\Delta \to 1$, and $C''(\Delta) > 0$. These conditions ensure that the optimal level of Δ is always less than one. We refer to the Δ chosen by an auditor as his "effort" level. We further assume that $C'''(\Delta) \geq 0$ in order that the equilibrium level of the auditor's effort is a well-behaved function of his expected liability.

Each period, the auditors compete for the client's engagement by submitting bids, F_i. Auditors' expected profits, and hence their bids, depend on their beliefs about the probability that management has a hidden risk because of the liability they incur if they fail to detect the risk and because the amount of effort they choose will depend on their assessment of client risk. In the first period, all auditors hold prior beliefs, ϕ°, about whether the client has a hidden risk. To avoid the possibility that a solvent auditor cannot compete with a wealth-constrained auditor, we assume that the wealth-constrained auditor has enough wealth to cover the expected liability, that is, $W \geq \phi^\circ L$. In the second period the client's current auditor (hereafter the incumbent and denoted by $i = I$) knows more about the probability of a hidden risk than the other auditors (hereafter the potential successors). Specifically, upon auditing the client, the incumbent gains private access to a signal and forms posterior beliefs ϕ. For simplicity and without loss of generality, we interpret ϕ as a private signal to the incumbent, with nondegenerate density $g(\phi)$ and c.d.f. $G(\phi)$.

We assume that the potential successor auditors bid first and the incumbent auditor responds to these bids. While this arrangement may not capture actual bargaining over audit fees, the assumption guarantees the existence of a pure strategy equilibrium. This assumption is by no means innocuous because, as we shall see, the pure strategy equilibrium we characterize does not hold if all auditors bid simultaneously. However, the assumed order of bidding does allow for a simple and intuitive characterization of the equilibrium bids which would not be possible using simultaneous bidding.

The equilibrium concept used in this model is the sequential equilibrium satisfying the Cho and Kreps [1987] Intuitive Criterion. This equilibrium is defined by the auditors' bid and effort choices and the client's auditor choice such that (*i*) the client's strategy is to choose the auditor who minimizes total expected auditing costs net of the benefit from hiring a solvent auditor (or equivalently, net of the opportunity cost of hiring a bad auditor); (*ii*) given the client's strategy and taking the bids of the other potential successors, F_{-i},[4] and the bidding *function* of the incumbent, $F_I(\phi)$, as fixed, each potential successor auditor i, knowing the effort he will take if hired, Δ_i, bids F_i to maximize his expected profit; (*iii*) given the client's strategy and knowing the successor auditors' bids, F_{-I}, the incumbent auditor, knowing the probability the client

[4] Let the subscript $-i$ mean all auditors other than auditor i. Hence, F_{-I} represents the set of all bids other than the incumbent auditor's.

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has a hidden risk, ϕ, and the amount of effort he will take if retained, Δ_I, chooses his bid, $F_I(\phi)$, to maximize his expected profit; (*iv*) the auditor i who wins the engagement chooses his effort, Δ_i, conditional on whether the first-period auditor resigned, to minimize the cost of effort plus expected liability; and (*v*) off-equilibrium beliefs satisfy the Intuitive Criterion. The remainder of this section characterizes the equilibrium strategies of the client, the incumbent auditor, and the potential successor auditors.

2.2 CLIENT'S SECOND-PERIOD STRATEGY

In the second period, the client chooses the auditor who minimizes his total expected audit cost (audit fee plus expected penalty) net of the opportunity cost of hiring a wealth-constrained auditor. Therefore, the client's strategy is to choose an auditor satisfying $i \in \arg\min_{\hat{i} \in A} (F_{\hat{i}} + \delta\Delta_{\hat{i}}P + \gamma_{\hat{i}}B)$ where δ and $\gamma_{\hat{i}}$ are indicator functions. δ is 1 if the client has a hidden risk (hereafter a "bad" client) and 0 otherwise (hereafter a "good" client), while $\gamma_{\hat{i}}$ is 1 if auditor $\hat{i} \in \mathbf{\Omega}$ and 0 if $\hat{i} \in \mathbf{\Sigma}$. The indicator function δ captures the assumption that no auditor infers a risk when none exists; γ incorporates the opportunity cost of hiring a wealth-constrained auditor.

For simplicity, this construction takes the benefit of hiring a solvent auditor as exogenous and independent of the circumstances under which that auditor was hired. More realistically, financial statement users may perceive the relative benefit of a solvent auditor differently if the auditor has been replaced than if the client simply retains the incumbent.[5]

Because his expected penalty is zero, a good client will switch auditors whenever a potential successor bids less, net of $\gamma_{\hat{i}}B$, than the incumbent. However, a bad client may retain the incumbent even though a potential successor bids less net of $\gamma_{\hat{i}}B$ if the probability of detection is higher with the potential successor auditor. The dependence of the client's strategy on his privately known type (i.e., the sorting property) generates the adverse selection which is critical to the characterization of an equilibrium supporting auditor resignations.

2.3 INCUMBENT AUDITOR'S SECOND-PERIOD STRATEGY

Given the client's second-period strategy and the potential successor auditors' bids, the incumbent auditor will choose a bid based on his knowledge of ϕ to maximize his expected profits. For each observed value of ϕ, the incumbent auditor's bid, $F_I(\phi)$, can be classified into one of four distinct strategy categories:[6]

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[5] While making the benefit of hiring a solvent auditor endogenous and/or dependent on the client firm's type would complicate the analysis, we do not believe it would change our results qualitaively.

[6] Here strategies are written for a solvent incumbent because this is the case which gives rise to resignations. For a wealth-constrained incumbent, simply replace L with W everywhere in these expressions.

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(*I*) Bid to retain the client regardless of his type

$$F_I(\phi) = \arg\max_{F_I}\{F_I - [1 - \Delta_I(\phi L)]\phi L - C(\Delta_I(\phi L))\}$$

subject to:

$$\Delta_I(\phi L) \in \arg\min_{\Delta\in[0,1]}\{(1-\Delta)\phi L + C(\Delta)\} \qquad (i)$$

$$F_I + E_\phi\,[\Delta_I\,(\phi L) \mid F_I,\ bad]P + \gamma_I B \le \min_{\hat{i}\in \mathbf{A}\backslash I}\{F_{\hat{i}} + \Delta_{\hat{i}}P + \gamma_{\hat{i}}B\} \qquad (ii)$$

$$F_I + \gamma_I B \le \min_{\hat{i}\in \mathbf{A}\backslash I}\{F_{\hat{i}} + \gamma_{\hat{i}}B\} \qquad (iii)$$

where $E_\phi\,[\Delta_I\,(\phi L)|\ F_I,\ bad]$ is a bad client's expectation of the incumbent auditor's effort, conditional on observing that the incumbent bids to retain the client.

(*II*) Bid to retain only a good client

$$F_I(\phi) = \arg\max_{F_I}(1-\phi)F_I$$

subject to:

$$\Delta_I = 0 \qquad (i)$$

$$F_I + \gamma_I B > \min_{\hat{i}\in \mathbf{A}\backslash I}\{F_{\hat{i}} + \Delta_{\hat{i}}P + \gamma_{\hat{i}}B\} \qquad (ii)$$

$$F_I + \gamma_I B \le \min_{\hat{i}\in \mathbf{A}\backslash I}\{F_{\hat{i}} + \gamma_{\hat{i}}B\} \qquad (iii)$$

(*III*) Bid to retain only a bad client

$$F_I\,(\phi) = \arg\max_{F_I}\phi\{F_I - [1 - \Delta_I(L)]L - C(\Delta_I(L))\}$$

subject to:

$$\Delta_I(L) \in \arg\min_{\Delta\in[0,1]}\{(1-\Delta)L + C(\Delta)\} \qquad (i)$$

$$F_I + \Delta_I(L)P + \gamma_I B \le \min_{\hat{i}\in \mathbf{A}\backslash I}\{F_{\hat{i}} + \Delta_{\hat{i}}P + \gamma_{\hat{i}}B\} \qquad (ii)$$

$$F_I + \gamma_I B > \min_{\hat{i}\in \mathbf{A}\backslash I}\{F_{\hat{i}}B + \gamma_{\hat{i}}B\} \qquad (iii)$$

(*IV*) Bid so that no client is retained, guaranteeing profit of zero, by choosing $F_I(\phi)$ to satisfy both

$$F_I + \Delta_I P + \gamma_I B > \min_{\hat{i}}\{F_{\hat{i}} + \Delta_{\hat{i}}P + \gamma_{\hat{i}}B\} \qquad (ii)$$

$$F_I + \gamma_I B > \min_{\hat{i}}\{F_{\hat{i}} + \gamma_{\hat{i}}B\} \qquad (iii)$$

In each of programs (*I*) through (*IV*) the objective function is the incumbent's expected profit from retaining the relevant client type(s) given his knowledge of ϕ. Constraint (*i*) in each of programs (*I*) through (*III*) is the sequential rationality constraint on the auditor's choice of effort, given the type(s) of client engaged. In program (*IV*), since no client is engaged, the incumbent auditor's effort choice is off equilibrium. This point is crucial to the characterization of an auditor resignation and we return to it later.

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Conditions (*ii*) and (*iii*) of each program are the sorting conditions required to retain the relevant client type(s), given the bids of the potential successors. Condition (*ii*) determines whether a bad client accepts the incumbent's bid and condition (*iii*) determines whether a good client accepts the incumbent's bid.[7]

The maximum of the solutions to programs (*I*) through (*IV*) determines the incumbent's response to the potential successor bids. This response separates client-initiated auditor switches from auditor resignations. When the incumbent uses a strategy from either category (*II*) or (*III*), some clients retain the incumbent while others do not, giving rise to client-initiated auditor switches. In contrast, the incumbent uses a strategy from category (*IV*) to *purposely drive away* clients. Specifically, when the incumbent assesses a high probability that his client is bad, he will protect himself from the possibility of retaining only a bad client by posting a bid so high that no client will accept it. For this reason, we refer to such a strategy as an auditor resignation. We have:

DEFINITION. Any bid made by the incumbent auditor which is higher than the cost of auditing *any* client and which is made in order to drive away all clients is said to be a resignation.

This definition distinguishes an active resignation from a noncompetitive (i.e., losing) bid. Empirically, the distinction is clear since a resignation is a refusal to bid. However, theoretically, a refusal to bid is equivalent to posting an infinitely high bid. So whether a bid is a resignation or just a noncompetitive bid depends on its level. Our definition of resignation suggests that any bid exceeding the cost of auditing the worst possible client is effectively infinite, and therefore theoretically equivalent to a resignation. This definition is meant to capture that, strategically, resignation is an intentional response which protects the incumbent from the possibility of losing money by having his bid accepted. As we shall see, active resignation in our model is *not* the result of an incumbent's cost disadvantage and inability to compete; it is the result of an adverse selection problem that only he faces.

A careful examination of programs (*II*) and (*III*) gives us our first result. We have:

LEMMA 1. When audit effort is chosen after the client is engaged, it is impossible to support an equilibrium in pure strategies where, for any observed ϕ, the incumbent auditor makes a separating bid. That is, the incumbent auditor never bids to retain only good clients or only bad clients in equilibrium.

Lemma 1 follows from the fact that programs (*II*) and (*III*) do not allow for any feasible auditor bids; given the sequential rationality requirement on auditor effort, it is impossible to satisfy both conditions (*ii*) and (*iii*) of either program.

[7] For convenience, we assume the client retains his incumbent auditor when indifferent.

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Given Lemma 1, the only possible equilibria involve the incumbent auditor, upon seeing ϕ, posting either a category (I) bid or a category (IV) bid. This allows us to qualitatively characterize the incumbent's equilibrium bidding strategy.

LEMMA 2. The incumbent auditor retains the client if and only if he observes the probability of a hidden risk to be sufficiently low.

As a result of Lemma 2, we can think of the incumbent's strategy as choosing a cutoff level of probability, $\phi^* \in [0,1]$, above which he makes a category (IV) bid and below which he makes a category (I) bid.

2.4 POTENTIAL SUCCESSOR AUDITORS' SECOND-PERIOD STRATEGIES

We assume there are at least two potential successor auditors who are solvent and two who are wealth constrained. Then price competition drives all potential successors' equilibrium bids to zero expected profit, allowing for a simple characterization of the potential successors' bids. The following lemma is used in characterizing the equilibrium potential successor bids.

LEMMA 3. A client that leaves a solvent incumbent auditor hires a wealth-constrained successor auditor.

Intuitively, solvent potential successors have the same cost of auditing the client as a solvent incumbent but are at an information disadvantage, so they cannot effectively compete for the client. On the other hand, a wealth-constrained auditor faces a *lower* audit cost, which may overcome the information disadvantage plus the benefit of hiring a solvent auditor, allowing a wealth-constrained successor to compete successfully against a solvent incumbent.

Taking Lemmas 2 and 3 together gives the zero expected profit bids of the wealth-constrained potential successors, F_W:[8]

$$F_W^* = (1 - \Delta_W^*) W \times E(\tilde{\phi} \mid \tilde{\phi} > \phi^*) + C(\Delta_W^*)$$

where:

$$\Delta_W^* \in \underset{\Delta \in [0,1]}{\arg\min} \{(1 - \Delta) W \times E(\tilde{\phi} \mid \tilde{\phi} > \phi^*) - C(\Delta)\}.$$

Recall that ϕ^* is the equilibrium cutoff level chosen by the solvent incumbent auditor. Since both good and bad clients hire wealth-constrained successors after leaving a solvent incumbent, a wealth-constrained successor's expectation about the client's type, conditional on the client accepting his bid, is $E(\tilde{\phi} \mid \tilde{\phi} > \phi^*)$.

[8] All wealth-constrained potential successors will bid the same amount because it is impossible to support an equilibrium in which good and bad clients are separated across wealth-constrained successors. This is because a bad client would choose a wealth-constrained successor who is bidding for a good client.

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3. *Equilibrium*

3.1 CHARACTERIZATION OF THE SECOND-PERIOD EQUILIBRIUM

We can now derive the second-period equilibrium when the first-period auditor is solvent. We emphasize the case of the solvent incumbent auditor because it gives rise to high-quality auditors resigning engagements, with the clients going to lower-quality auditors. Once this is done, we still must verify that the conditions which give rise to auditor resignation in Proposition 1 are consistent with conditions that support the initial engagement of a solvent auditor.

We characterize the sequential equilibria of our model and further restrict attention to the equilibrium which satisfies the Intuitive Criterion of Cho and Kreps [1987]. This refinement is specifically used to show how the incumbent benefits from resigning an audit engagement, rather than trying to price out an increase in expected liability. As such, this refinement technique provides a formal distinction between a resignation and a noncompetitive bid.

PROPOSITION 1. A solvent incumbent auditor will retain the client when the likelihood of a hidden risk is sufficiently low. However, the incumbent will resign the audit when the likelihood of a hidden risk is sufficiently high, provided the benefit of hiring a solvent incumbent is not too high. After a resignation, the client's successor auditor will be wealth constrained. Specifically,

$$F_S^* \geq (1 - \Delta(L))L + C(\Delta(L))$$

where $\Delta(L)$ satisfies $C'(\Delta(L)) \equiv L$.

$$F_W^* = (1 - \Delta_W^* W \times E(\tilde{\phi} \mid \tilde{\phi} > \phi^*) + C(\Delta_W^*),$$

where Δ_W^* satisfies $C'(\Delta_W^*) \equiv W \times E(\tilde{\phi} \mid \tilde{\phi} > \phi^*)$.

$$F_I^* = \begin{cases} F_W^* + B & \text{when } \phi \leq \phi^* \\ \text{“}resign\text{”} & \text{when } \phi > \phi^* \end{cases}$$

where ϕ^* is the lowest value of ϕ satisfying:

$$\phi \equiv \frac{(1-\Delta_W)\, W \times E(\tilde{\phi} \mid \tilde{\phi} > \phi^*) + C(\Delta_W) - (C\Delta_I(\phi L)) + B}{(1-\Delta_I(\phi L))\, L}$$

and Δ_W satisfies $C'(\Delta_W) \equiv W \times E(\tilde{\phi} \mid \tilde{\phi} > \phi)$.

Proposition 1 indicates that when a solvent incumbent auditor finds that the probability his client has hidden risk is sufficiently high, he will always resign the audit, leaving the client with a wealth-constrained auditor. As we shall see, if the incumbent attempts to "price out" the increase in expected liability associated with his revised beliefs by bidding

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$(1 - \Delta_I(\phi L))\phi L + C(\Delta_I(\phi L))$ when he sees $\phi > \phi^*$, only bad clients may choose to retain him. To avoid a loss, the incumbent's bid must drive away a bad client. Such a bid is strategically equivalent, from the incumbent's perspective, to resigning from the engagement, refusing to make any bid at all.

Clearly the incumbent loses his cost advantage (and therefore the client) when the probability that the client is bad is sufficiently high. However, it is less clear why the outcome is an active resignation rather than a price increase to cover the respected costs. The answer lies in the incumbent's off-equilibrium beliefs about the type of client that would accept a bid equal to the auditor's expected cost. If the incumbent prices out his expected cost by bidding $F_I = (1 - \Delta_I(\phi L))\phi L + C(\Delta_I(\phi L))$ when ϕ exceeds the cutoff level ϕ^*, the bid would exceed $F_W^* + B$, so the client would switch auditors in equilibrium. But for this bid to be sequentially rational, it must generate nonnegative expected profit in the off-equilibrium event that the client does not switch. Hence, to support this bid in a sequential equilibrium, the incumbent must hold off-equilibrium beliefs that a good client is at least as likely as a bad client to accept this bid. Otherwise the bid generates an expected loss and is dominated by resignation. We argue that such off-equilibrium beliefs are unreasonable using the Intuitive Criterion of Cho and Kreps [1987] and, therefore, that bidding expected cost when $\phi > \phi^*$ is not an equilibrium response for the incumbent.

Under the Intuitive Criterion, off-equilibrium beliefs must put zero weight on the probability a player type will take an action that gives lower expected payoff than his equilibrium payoff. In our setting, this means that an incumbent who bids $F_I > F_W^* + B$ must believe that a client who accepts his bid (off equilibrium) must be bad. While a good client cannot possibly benefit from accepting the incumbent's bid, a bad client who believes Δ_I is sufficiently low relative to Δ_W would. As a result, the off-equilibrium beliefs of the incumbent necessary to support an equilibrium bid of $F_I = (1 - \Delta_I(\phi L))\phi L + C(\Delta_I(\phi L))$ when $\phi > \phi^*$ fail the test of the Intuitive Criterion. In fact, the only allowable off-equilibrium beliefs under this refinement imply that the incumbent's bid be greater than or equal to the cost of auditing a known bad client, $(1 - \Delta_I(L))L + C(\Delta_I(L))$. Any such bid is equivalent to an auditor resignation, by our definition.

The existence of two distinct types of client is crucial to our characterization of an auditor resignation. If there is only one type of client, the incumbent auditor's off-equilibrium beliefs about the client's type are necessarily the same as his on-the-equilibrium-path beliefs. This means that his bid of expected cost is sequentially rational because it also generates zero expected profit given his off-equilibrium beliefs. Active resignation is the product of an adverse selection problem faced by the incumbent, and adverse selection requires the existence of more than one type of client.

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Also crucial to the bidding strategies in Proposition 1 is the assumed order of bidding. If the incumbent auditor was not allowed to react to the successor bids, the successor bids are not a best response. Taking the incumbent's bid, F_I^*, as fixed, a successor can bid slightly less and thereby obtain the client. As a result, no pure strategy bidding equilibrium exists when the auditors all bid in ignorance of each other's bids.

Proposition 1 takes for granted the solvency of the incumbent auditor. We still must verify that the conditions which give rise to auditor resignation in Proposition 1 are consistent with conditions that support the initial engagement of a solvent auditor.

3.2 CHARACTERIZATION OF THE FIRST-PERIOD EQUILIBRIUM

In this section we establish conditions under which a client hires a solvent auditor in the first period, only to have the auditor resign in the second period. This establishes the plausibility of Proposition 1.

LEMMA 4. There always exist levels of B and ϕ° such that a solvent auditor wins the engagement in the first period but resigns in the second period if he discovers that the probability that the client is bad is sufficiently high.

The client will hire a solvent auditor as long as the benefit of doing so exceeds the two-period expected cost advantage of the wealth-constrained auditor. But once a solvent auditor is hired, resignation depends on the wealth-constrained auditor's cost advantage given a realization of ϕ. Regardless of the *expected* second-period cost advantage, there exist realizations of ϕ such that the cost advantage of a wealth-constrained auditor overcomes the benefit B of hiring a solvent auditor, provided B is not too large.

The following alternative interpretation of our model also rationalizes second-period resignations, without the need to restrict attention to a subset of economies. The results of Proposition 1 depend only on the difference in liability faced by the solvent and wealth-constrained auditors, regardless of its source. For example, the difference could occur because auditors who fail to detect a hidden risk are held to a "market model" of damages in which the auditor must pay the difference between the amount investors paid for the firm and its value after the fraud is revealed. When an auditor resignation is viewed by investors as indicative of risk, the price of the firm would fall following a resignation. Therefore, the liability incurred by an auditor who retains his client all the way through the eventual consequence of a hidden risk, without ever exposing that risk, would be higher than the liability of a successor who obtained the client after the first auditor had resigned (presumably under the suspicion of a hidden risk). Thus, the auditors can be identical in the first period, and resignation is still always possible in the second period, because potential successors always face lower liability than incumbents.

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4. *Comparative Statics*

In this section we study the effect of increased auditor liability on the characteristics of the audit market equilibrium. Our first comparative static substantiates auditors' claims that increased liability causes an increase in the frequency of auditor resignations.[9]

PROPOSITION 2. When the incumbent auditor is solvent, an increase in auditor liability will increase the frequency of auditor resignations.

Increasing L increases the expected auditing cost only for the solvent auditors, thereby making it more difficult for them to compete with wealth-constrained potential successors; the result is more frequent resignations. Of course Proposition 2 is a local statement, because a sufficiently large increase in L means solvent auditors would never engage a client in the first place. Hence there would be no resignations in the second period. To interpret Proposition 2 as a global statement, the conditions of Lemma 4 must continue to hold upon increasing L.

In seeking liability relief, auditors have argued that increases in liability have increased the frequency of resignations which are in turn detrimental because clients are forced to turn to inferior auditors (Holland, Galen, and Light [1993]). These arguments have sometimes been met with skepticism and suggestions that the incumbent auditors could simply price out the increase in their expected costs rather than resign (e.g., Gode [1993]). Proposition 2 rationalizes the auditors' claims. However, it is not yet clear from this model that resignations harm anyone other than the incumbent auditor. In particular, for the marginal client type, ϕ^*, the incumbent would choose effort based on knowledge of ϕ^*. But the successor's information disadvantage increase the uncertainty he faces about the client's type; the successor chooses effort based only on knowledge that $\phi > \phi^*$. The extra risk he assumes makes the benefit of effort greater for the successor, so resigned clients might actually face *greater* probability of detection by successor auditors. If so, increased resignation could increase the overall frequency of risk detection. The next proposition outlines the effect of increased auditor liability and increased auditor resignation on the *overall* frequency of risk detection.

PROPOSITION 3. When the incumbent auditor is solvent, an increase in auditor liability will increase the frequency of risk detection among the clients the incumbent retains but will decrease the frequency of risk detection among the clients for which the auditor resigns.

Proposition 3 shows that, while increased auditor resignation is detrimental in that it decreases the frequency of risk detection, increased auditor liability can be beneficial because it can increase the overall probability of detecting risk.[10] However, even if maximizing the fre-

[9] When we change L in the following analysis, we continue to assume that $L \geq W$ and $L \geq P$.

[10] An important caveat is that the presence of hidden risk is exogenous.

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quency of risk detection is the social objective, increasing liability is not necessarily beneficial. The result is similar to Dye's [1993] result that increasing auditing standards can decrease the average quality of audits. This is formalized in the following proposition.

PROPOSITION 4. The overall frequency of risk detection in the economy may decrease in the level of auditor liability.

Proposition 4 can be compared to existing theories of liability in which it is generally understood that potential injurers can be led to act in a socially optimal way by increasing liability to the point where any social inefficiency is eliminated, except when the potential injurer is wealth constrained (see, for example, Shavell's [1987] discussion). If maximizing the frequency of risk detection is our social objective, Proposition 4 shows that increasing liability does not necessarily achieve this goal. This is because, in our setting, increasing liability increases the cost advantage of the wealth-constrained auditors but does not directly affect their effort incentives. So the increase in clients seeking audits from wealth-constrained auditors can result in decreased overall detection when liability increases. The general insight that increasing liability can result in less socially desirable behavior because it makes wealth-constrained producers more competitive may be useful in other contexts, especially where producers are truly potential injurers.

5. *Conclusion*

This paper presents a model which rationalizes the claims that increased auditor liability leads to more auditor resignation and lower-quality audits. The model shows that auditor resignations arise when there are both potential successor auditors who face lower liability than the incumbent auditor and client firms who privately know whether they have a hidden risk.

The incentives of the liability system increase the frequency of risk detection among clients the incumbent auditor retains, but these are the *least* likely to have a hidden risk. We show that the increase in resignations combined with the decrease in risk detection among resigned clients can mean that large increases in liability actually reduce the overall frequency of risk detection.

We made several important simplifying assumptions to obtain these results. First, we assume the presence (or absence) of hidden risk is exogenous. In general, however, one would expect that this risk would be influenced by the level of auditor liability. Also, while we do not explicitly model a capital market which reacts to auditor resignations, we do include an exogenous benefit associated with hiring a solvent auditor. Again, we would expect the capital market reaction to a resignation to affect the client's auditor choice.

Notwithstanding these caveats, the model yields several empirical implications. We predict that following a resignation, the client will engage a more wealth-constrained auditor. And since auditor resignations

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occur when the incumbent auditor believes it is relatively likely that the client has a hidden risk, we would expect that firms whose auditors resign have a higher incidence of adverse outcomes than other firms. We additionally predict that, consistent with anecdotal evidence and auditors' claims, the incidence of resignations will increase with the level of auditor legal liability. Finally, our model predicts a decline in a client firm's market value following an auditor resignation since resignation indicates an increase in the probability of a hidden risk.

APPENDIX A

Proof of Lemma 1. Because $\Delta_i \geq 0$, conditions (*ii*) and (*iii*) of program (*II*) cannot be satisfied simultaneously. Similarly, if $\Delta_I(L) \geq \Delta_i$ for all $i \neq I$, conditions (*ii*) and (*iii*) of program (*III*) cannot be satisfied simultaneously. To complete the proof, we show that $\Delta_I(L) \geq \Delta_i$ for all $i \neq I$. Equilibrium requires that the hired auditor chooses his effort to minimize his total expected auditing cost. Note that, regardless of an auditor's type, his total cost can be defined as $[1 - \Delta(\theta)]\theta + C(\Delta(\theta))$, where θ is the auditor's assessment of the probability the client is bad times the penalty applicable to the auditor. The first-order condition on the auditor's choice of effort gives $C'(\Delta(\theta)) \equiv \theta$. While the assumption that $C''(\Delta) > 0$ implies through implicit differentiation of the first-order condition that $\Delta'(\theta) > 0$. Finally, noting that the maximum expected liability to any auditor is $\theta = L$ gives $\Delta_I(L) \geq \Delta_i$ for all $i \neq I$. *Q.E.D.*

Proof of Lemma 2. Since resignation guarantees the incumbent auditor zero profit, he will choose to retain the client iff posting a bid that satisfies the conditions of program (*I*) generates positive expected profit. Note that either condition (*ii*) or condition (*iii*) of program (*I*) will be binding and consequently define the incumbent auditor's bid. Suppose condition (*iii*) is binding. Then the incumbent auditor, upon observing ϕ, will bid to retain the client iff:

$$[1 - \Delta(\theta)]\theta + C(\Delta(\theta)) \leq \min_i \{F_i + \gamma_i B\} - \gamma_I B \tag{1}$$

where $\theta \equiv \phi L$.

Given the potential successors' bids, the right-hand side of (1) is constant in ϕ. Differentiating the left-hand side and applying the envelope theorem shows that the change in total cost with respect to θ is $(1 - \Delta(\theta))$, which is greater than zero. Noting that θ is increasing in ϕ proves that (1) is satisfied iff ϕ is sufficiently small. An identical argument applies to the case of condition (*ii*) binding, thus proving the lemma. *Q.E.D.*

Proof of Lemma 3. By means of contradiction, suppose that an equilibrium exists in which a client leaves a solvent incumbent to hire a solvent successor. Note that such an equilibrium would require the incumbent to adopt a category (*IV*) bid and therefore that:

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$$F_I < [1 - \Delta(\phi L)]\phi L - C(\Delta(\phi L)). \tag{2}$$

If the client is good, such an equilibrium would also require that:

$$F_I > \min_i \{F_{\hat{i}} + \gamma_{\hat{i}} B\} = F_S \tag{3}$$

where F_S is the successful bid of a solvent successor auditor.

Note that (2) and (3) together imply that the solvent successor's bid, F_S, generates negative expected profit conditional on its acceptance. Therefore, there are no equilibria in which a good client leaves a solvent successor to hire a solvent incumbent. We still need to check for the possibility that only bad clients hire solvent successors after leaving a solvent incumbent. In addition to condition (2), such an equilibrium would require that:

$$F_I + \Delta_I P > \min_i \{F_{\hat{i}} + \gamma_{\hat{i}} B + \Delta_{\hat{i}} P\} = F_S + \Delta(L)P. \tag{4}$$

Recall from the proof of Lemma 2 that $\Delta(L) \geq \Delta_i$ for all i. Therefore, satisfaction of condition (4) implies $F_I > F_S$. And condition (2) then once again implies that F_S generates negative expected profit, contradicting its use as an equilibrium bid. *Q.E.D.*

Proof of Proposition 1. By Lemmas 1 and 2, the incumbent will bid to retain the client regardless of his type (category (I)) when $\phi \leq \phi^*$ and bid so that no client is retained (category (IV)) when $\phi > \phi^*$. First we examine a solvent incumbent's optimal category (I) bid.

CLAIM. Constraint (ii) of program (I) is sufficient for constraint (iii) and is therefore binding, giving $F_I^*(\phi) = F_W^* + B$ whenever $\phi \leq \phi^*$.

Proof of Claim. Because the successor learns that the incumbent did not resign (and therefore must have used a category (I) bid) after he bids but before he chooses Δ, Δ_W in program (I) is off equilibrium. The Intuitive Criterion requires that the successor hold off-equilibrium beliefs of $\phi = 1$ conditional on the client accepting his bid when the incumbent does not resign, and hence that $\Delta_W = \Delta(W)$. To complete the proof of the claim we show that $\Delta_W \geq E_\phi[\Delta_I(\phi L) \mid F_I, bad]$, because then satisfaction of constraint (ii) implies satisfaction of constraint (iii). The assumption that $W \geq \phi^\circ L$ and the fact that $\Delta'(\theta) > 0$ implies $\Delta(W) \geq \Delta(\phi^\circ L)$. And $\Delta(\phi^\circ L) \geq \Delta(E(\phi \mid \phi \leq \phi^*)L)$ because $\phi^* \leq 1$. The assumption that $C'''(\Delta) \geq 0$ implies $\Delta''(\theta) \leq 0$ and therefore that $E_\phi[\Delta_I(\phi L) \mid \phi \leq \phi^*] \leq \Delta(E(\phi \mid \phi \leq \phi^*)L)$. Finally, since in equilibrium $E_\phi[\Delta_I(\phi L) \mid F_I, bad] \leq E_\phi[\Delta_I(\phi L) \mid \phi \leq \phi^*]$ whenever $F_I \leq F_W + B$, $\Delta_W \geq E_\phi[\Delta_I(\phi L) \mid F_I, bad]$.

Given Lemmas 1 and 2, the equilibrium bids F_I^* and F_W^*, and the fact that the incumbent uses a category (I) bid iff it generates positive expected profit, the equilibrium cutoff value, ϕ^*, must satisfy:

$$[1 - \Delta_I(\phi^* L)]\phi^* L + C(\Delta_I(\phi^* L)) \equiv$$
$$[1 - \Delta(E(\tilde{\phi} \mid \tilde{\phi} > \phi^*)W)]E(\tilde{\phi} \mid \tilde{\phi} > \phi^*)W + C(\Delta(E(\tilde{\phi} \mid \tilde{\phi} > \phi^*)W)) + B. \tag{5}$$

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Rearranging (5) gives the expression in Proposition 1.

At $\phi = 0$, the right-hand side of (5) is greater than the left-hand side of (5). At $\phi = 1$, the left-hand side of (5) is greater than the right-hand side of (5) if and only if:

$$[1 - \Delta(L)]L + C(\Delta(L)) > [1 - \Delta(W)]W + C(\Delta(W)) + B. \qquad (6)$$

If (6) holds, as is the case for B not too large, the intermediate value theorem gives existence of a $\phi^* < 1$ satisfying (5).

CLAIM. If expression (5) is satisfied for more than one value of ϕ^*, it is the lowest such value which uniquely determines the equilibrium.

Proof of Claim. Let ϕ' be the smallest value of ϕ^* satisfying (5) and suppose that there exists another level of ϕ^* satisfying (5), say $\phi'' > \phi'$. Let the associated equilibrium wealth-constrained successor bids be F_W'' $> F_W'$. For ϕ'' to support an equilibrium, it must be that all wealth-constrained auditors are bidding F_W''. If a wealth-constrained auditor was to defect from the proposed equilibrium and bid $F_W' + \varepsilon$, the incumbent, acting as a Stackelberg follower, would respond with $F_I = F_W' + \varepsilon + B$ and would choose to resign iff $\phi > \hat{\phi}$, where $\hat{\phi}$ satisfies $[1 - \Delta(\hat{\phi}L)]\ \hat{\phi}L + C(\Delta(\hat{\phi}L)) = F_W' + \varepsilon + B$. Note that for all $\phi > \phi'$ in a nbd(ϕ'), $[1 - \Delta(E(\tilde{\phi} \mid \tilde{\phi} > \phi)W)]\ E(\tilde{\phi} \mid \tilde{\phi} > \phi)W + C(\Delta(E(\tilde{\phi} \mid \tilde{\phi} > \phi)W)) + B < [1 - \Delta(\phi L)]\ \phi L + C(\Delta(\phi L))$. Then there exists an $\varepsilon > 0$ such that the $\hat{\phi}$ satisfying $[1 - \Delta(\hat{\phi}L)]\ \hat{\phi}L + C(\Delta(\hat{\phi}L)) = F_W' + \varepsilon + B$ is greater than ϕ' and in a nbd(ϕ') so that $F_W' + \varepsilon > [1 - \Delta(E(\tilde{\phi} \mid \tilde{\phi} > \hat{\phi})W)]\ E(\tilde{\phi} \mid \tilde{\phi} > \hat{\phi})W + C(\Delta(E(\tilde{\phi} \mid \hat{\phi} > \hat{\phi})W))$, generating strictly positive expected profit to the wealth-constrained successor. So F_W'' is not a best response for a wealth-constrained auditor and, therefore, the cutoff level ϕ'' cannot be supported in equilibrium.

Since the ϕ^* satisfying (5) is unique, it follows that the lowest bid satisfying (5), F_W^*, determines the unique equilibrium cutoff value, ϕ^*. We now establish that the solvent incumbent auditor actively resigns (as we define it) when ϕ is greater than ϕ^*. The Intuitive Criterion requires the incumbent's beliefs put probability 1 on the client being fraudulent in the off-equilibrium event of the client accepting an incumbent bid $F_I > F_W + B$. In addition, all bids must generate nonnegative expected profit at every information set, even those off equilibrium. Therefore, for a bid $F_I > F_W + B$ to satisfy the Intuitive Criterion, it must be greater than or equal to $[1 - \Delta(L)]L + C(\Delta(L))$. Any such bid satisfies our definition of resignation.

Finally, we characterize the solvent potential successors' equilibrium bids. Note from the proof of Lemma 2 that bids that will not generate expected losses must satisfy $F_S \geq F_I$. But then the Intuitive Criterion requires that the solvent successor hold beliefs $\phi = 1$ conditional on his bid being accepted. Sequential rationality then requires $F_S \geq [1 - \Delta(L)]L + C(\Delta(L))$. *Q.E.D.*

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Proof of Lemma 4.

$$B > E\{[1 - \Delta_L(\phi L)]\phi L + C(\Delta_L(\phi L))\} - E\{[1 - \Delta_W(\phi W)]\phi W + C(\Delta_W(\phi W))\} + \{[1 - \Delta_L(\phi^\circ L)]\phi^\circ L + C(\Delta_L(\phi^\circ L))\} - \{[1 - \Delta_W(\phi^\circ W)]\phi^\circ W + C(\Delta_W(\phi^\circ W))\} \quad (7)$$

is sufficient for a solvent auditor to win the first-period engagement. Recall that condition (6) in the proof of Proposition 1 is a sufficient condition for the possibility of auditor resignation in the second period with a solvent incumbent. Rearranging (6) gives:

$$B < \{[1 - \Delta_L(L)]L + C(\Delta_L(L))\} - \{[1 - \Delta_W(W)]W + C(\Delta_W(W))\}. \quad (8)$$

Since the right-hand side of expression (7) is continuous in ϕ° and equal to zero at $\phi^\circ = 0$, and the right-hand side of expression (8) is strictly greater than zero for all ϕ°, there always exist levels of B satisfying both (7) and (8) when ϕ° is sufficiently small. *Q.E.D.*

Proof of Proposition 2. To prove that resignations increase in L recall that, from Proposition 1, ϕ^* is the smallest value of ϕ that satisfies:

$$[1 - \Delta_L(\phi L)]\phi L + C(\Delta_L(\phi L)) = (1 - \Delta_W)E[\tilde{\phi} \mid \tilde{\phi} > \phi]W + C(\Delta_W) + B. \quad (9)$$

At $\phi = 0$, the left-hand side of (9) is smaller than the right-hand side. Therefore, at ϕ^*, the derivative of the left-hand side of (9) with respect to ϕ must be weakly greater than the derivative of the right-hand side with respect to ϕ. So, differentiating both sides of (9) with respect to ϕ, applying the envelope theorem, and evaluating at ϕ^* gives us:

$$[1 - \Delta_L(\phi^* L)]L \geq (1 - \Delta_W^*)W \frac{\partial E[\tilde{\phi} \mid \tilde{\phi} > \phi^*]}{\partial \phi^*}. \quad (10)$$

Substituting ϕ^* into (9) defines an identity which, through implicit differentiation with respect to L and application of the envelope theorem, gives us:

$$\frac{d\phi^*}{dL} = -\frac{[1 - \Delta_L(\phi^* L)]\phi^*}{[1 - \Delta_L(\phi^* L)]L - (1 - \Delta_W^*)W \dfrac{\partial E[\tilde{\phi} \mid \tilde{\phi} > \phi^*]}{\partial \phi^*}}.$$

Finally, note that (10) implies $\dfrac{d\phi^*}{dL} < 0$. *Q.E.D.*

Proof of Proposition 3. From (9) we know that:

$$[1 - \Delta_L(\phi^* L)]\phi^* L + C(\Delta_L(\phi^* L)) > (1 - \Delta_W^*)E[\tilde{\phi} \mid \tilde{\phi} > \phi^*]W + C(\Delta_W^*). \quad (11)$$

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Recall from the proof of Lemma 2 that an auditor's total expected auditing cost is increasing in his expected liability. Therefore, (11) implies $L\phi^* > W \times E[\tilde{\phi} \mid \tilde{\phi} > \phi^*]$. Also recall that $\Delta'(\theta) > 0$, so $\Delta_L(\phi^* L) > \Delta^*_W$. Noting that $\Delta_L(\phi L) > \Delta_L(\phi^* L) > \Delta^*_W$ for all $\phi > \phi^*$, we see that resigned clients face lower detection probabilities than they would have had the auditor not resigned. Since increasing L increases resignations, it decreases detection among newly resigned clients. Furthermore, since Δ^*_W is increasing in ϕ^*, increasing L decreases detection among all resigned clients. However, since $\Delta_L(\phi L)$ is increasing in L, the probability of detection increases among the clients which the auditor retains when L increases. *Q.E.D.*

Proof Proposition 4. This proposition follows trivially from Proposition 3 by choosing a density, $g(\phi)$, which sufficient mass in the neighborhood of ϕ^*.

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